

Technical Progress Report: Time Prediction for Embryo Development Using Diffusion Models

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I. OVERVIEW

This project investigates time prediction of human embryo development using conditional diffusion models. Given a short temporal context of embryo frames ending at time t , the objective is to synthesize the subsequent frame at $t + 1$ and recursively generate future frames. The focus is on modeling realistic morpho-kinetic evolution in time-lapse microscopy data, rather than single-image reconstruction.

The approach follows a structured pipeline consisting of automated data preprocessing, auxiliary classification models for filtering and analysis, and a context-conditioned diffusion model trained on strictly adjacent frame pairs.

A. Project Goal

The primary goal is to develop and evaluate a diffusion-based model capable of predicting the next developmental state of a human embryo from a short history of observed frames. The model operates on preprocessed time-lapse sequences and is conditioned on a fixed-size temporal context (currently $k = 7$), with use of absolute frame index information for elapsed developmental time. Emphasis is placed on temporal consistency, visual realism, and alignment with known developmental phases.

II. DATA AND PREPROCESSING

A. Dataset Summary

The project uses a public time-lapse dataset of human embryos [1], where each embryo is recorded as a sequence of microscopy frames and accompanied by a CSV file describing morpho-kinetic phase transitions. Frames are acquired at fixed temporal intervals, enabling strict temporal ordering and adjacent-frame supervision. Each embryo is treated as an independent sequence throughout preprocessing and training.

B. Data Organization and Conventions

Raw frames are stored per embryo in individual folders, while phase annotation CSV files are stored separately. The preprocessing pipeline outputs a *silver* dataset organized by focal plane and embryo identifier. All output frames preserve their original 1-based frame indices. Frames removed during preprocessing are not renumbered, resulting in explicit index gaps that reflect filtered or missing time points.

Phase annotation CSV files are rewritten during preprocessing to remain consistent with the retained frame indices after filtering. This ensures alignment between image data and temporal annotations without introducing artificial continuity.

This convention preserves temporal integrity and avoids introducing implicit assumptions about missing or interpolated frames.

C. Embryo Presence Filtering

A binary embryo presence classifier is used to filter out frames that do not contain a visible embryo. The classifier is implemented as a convolutional neural network trained on grayscale embryo images and operates at the frame level.

During preprocessing, frames are first subsampled according to a fixed temporal stride and then independently evaluated by the presence classifier. Frames predicted as empty, corrupted, or lacking a visible embryo are discarded. This step removes background-only frames and prevents downstream models from learning spurious visual patterns unrelated to embryo morphology.

The trained presence model is loaded by the preprocessing pipeline and applied uniformly across all embryos and focal planes.

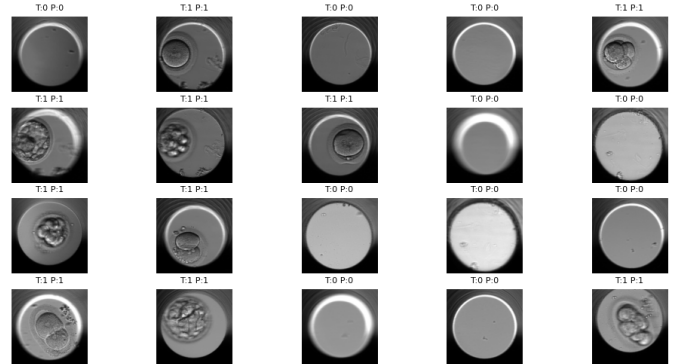


Fig. 1. Example frames containing embryos ($T = 1$) and frames without a visible embryo ($T = 0$), together with the predictions of the presence classifier ($P = 0, 1$).

D. Preprocessing Pipeline

The preprocessing pipeline is implemented as a single automated and deterministic process. For each embryo and focal plane, frames are processed in temporal order and undergo the following operations:

- **Presence filtering:** frames predicted as not containing an embryo are removed.
- **Spatial cropping:** a circular crop centered on the embryo is applied using a Hough-based detection method. If circle detection fails, a fallback center crop is used.
- **Resizing:** cropped frames are resized to a fixed spatial resolution to ensure consistent input dimensions.
- **Contrast normalization:** Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied to reduce illumination variability and enhance morphological detail.

Processed frames are written to disk using a consistent image format while preserving their original frame indices. Phase annotation CSV files are rewritten to reflect the retained frame indices after filtering and cropping.

No temporal interpolation, smoothing, or reindexing is performed. The pipeline can be rerun reproducibly without requiring changes to downstream training or evaluation code.

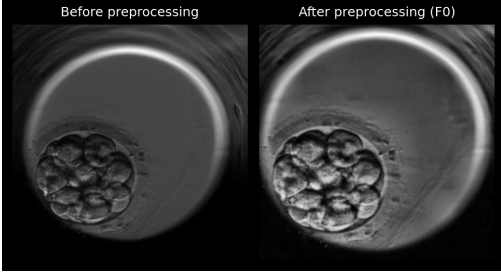


Fig. 2. Example frame before and after preprocessing, illustrating embryo-centered cropping and contrast normalization.

E. Focal Plane Handling

Preprocessing is performed independently for each focal plane present in the raw data. The pipeline supports multiple focal planes without modification, producing a separate *silver* dataset for each plane.

Unless otherwise stated, experiments reported in this work use a single focal plane to reduce modeling complexity and isolate temporal prediction performance.

F. Quality Control and Failure Cases

Basic consistency checks are applied during preprocessing, including verification of file readability and frame–annotation alignment. Frames that fail to load or produce invalid preprocessing outputs are skipped.

Misclassifications by the presence model or failures in embryo detection may result in short temporal gaps. These gaps are explicitly preserved and handled during dataset indexing and training pair construction rather than corrected heuristically, ensuring that temporal uncertainty is not artificially smoothed.

III. AUXILIARY MODELS (PRESENCE / PHASE)

Two auxiliary classification models are used in the pipeline: an embryo presence classifier and a morpho-kinetic phase classifier. These models are not part of the diffusion training objective. The presence classifier supports data preparation. The phase classifier supports analysis; classifier-guidance sampling was explored but is disabled in the final pipeline.

A. Embryo Presence Classifier

The embryo presence classifier is a binary convolutional model trained to determine whether a given frame contains a visible embryo. It operates on single grayscale frames and is applied independently to each time point. The model is trained

using manually curated examples of valid and invalid frames and is used exclusively during preprocessing.

During dataset construction, frames predicted as not containing an embryo are discarded. This filtering step removes empty wells, acquisition artifacts, and severely degraded frames.

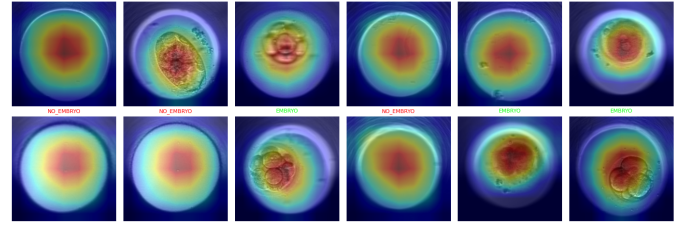


Fig. 3. Gradcam output of frames classified as containing an embryo and frames to be discarded.

B. Morpho-Kinetic Phase Classifier

A multi-class phase classifier is trained to map a single embryo frame to one of 16 morpho-kinetic phases. The classifier is trained using frames extracted from the silver dataset and labels derived from the provided phase annotation CSV files. Frames are sampled within annotated phase intervals, and class imbalance is mitigated using weighted sampling.

The model follows a convolutional architecture with grayscale inputs and incorporates timestep-dependent noise during training to improve robustness. Standard data augmentations are applied during training, while validation and test sets use deterministic preprocessing.

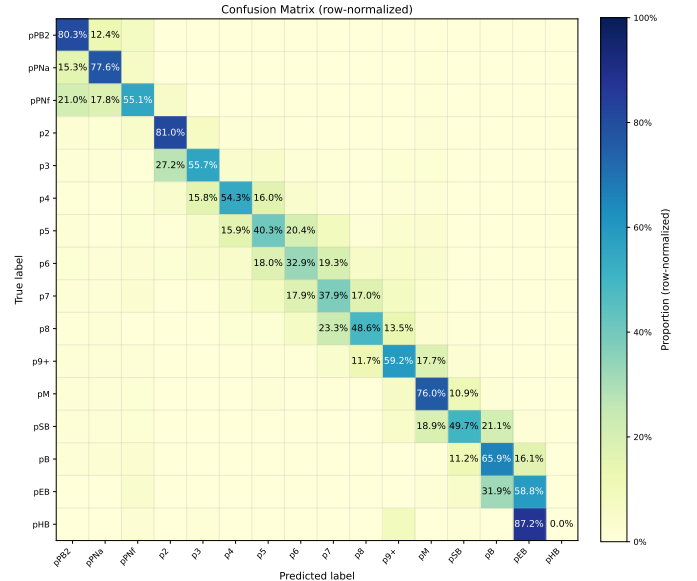


Fig. 4. Confusion matrix of the phase classifier evaluated on the test set.

C. Role in the Diffusion Pipeline

The presence classifier is a mandatory component of the preprocessing pipeline and directly affects the temporal structure of the training data. In contrast, the phase classifier is not

used during diffusion model training. Its outputs are used only for analysis.

The diffusion model learns temporal dynamics directly from visual data, without being constrained by explicit phase supervision during training.

Model	Used in Training	Used in Sampling
Presence classifier	Yes (preprocessing)	No
Phase classifier	No	No (analysis only)

TABLE I

SUMMARY OF AUXILIARY MODEL USAGE IN THE PIPELINE.

IV. DIFFUSION MODEL

A. Task Definition

The diffusion model is trained for next-frame prediction on the silver dataset (F0 only for now). Given the last k context frames ending at time t , the model generates the next frame at $t + 1$. Multi-step prediction is supported by rolling the context window forward and feeding generated frames back as new context.

Training is based on noise-prediction (DDPM objective), while sampling uses DDIM for efficiency.

B. Indexing and Data Splits

A cached frame index is built by scanning the silver dataset and pairing each frame with a phase label derived from the corresponding `embryo_id_phases.csv`. Frame numbers are parsed from filenames and treated as 1-based indices. Only frames that can be assigned a valid phase label are included in the index.

Train/validation/test splits are performed by embryo ID to prevent leakage across sets. Splits are cached to disk. Stratification uses the dominant phase label per embryo to maintain similar label distributions across splits.

Split	Embryos
Train	565
Val	69
Test	69

TABLE II

EMBRYO-LEVEL SPLIT SIZES (CACHED).

C. Pair Construction ($\text{Context} \rightarrow \text{Target}$)

Training samples are constructed as deterministic, overlapping windows. A valid window is defined by:

$$[x_{t-k+1}, \dots, x_t] \rightarrow x_{t+1},$$

where all context frames and the adjacent target x_{t+1} must exist. Missing frames lead to skipped windows rather than reindexing.

For validation and testing, windows are additionally categorized as *transition* or *non-transition* depending on whether the phase label changes between t and $t + 1$. This is used for targeted sampling and analysis.



Fig. 5. Context window construction for next-frame prediction: k previous frames predict $t + 1$.

D. Conditioning

The denoising network is conditioned on: (i) the stack of k context frames, and (ii) an absolute frame index embedding.

Context conditioning is implemented through cross-attention: the context stack is encoded into a token sequence, and the UNet attends to these tokens while predicting noise on the target. The absolute target frame index is embedded and projected to a scalar, then injected as a spatial *time map* conditioning channel.

A compact summary of the context stack is also provided to the UNet by projecting the k context channels to a single image-like channel via a 1×1 convolution, which is concatenated with the noisy target (and the time map if enabled).

E. Architecture

The model consists of a context encoder and a UNet denoiser:

- **Context encoder:** a small CNN that encodes each context frame into a spatial feature grid. Positional embeddings are added for (a) the context-frame order and (b) spatial token positions. Optionally, absolute frame-index embeddings are added to the per-frame context tokens.
- **UNet denoiser:** a `UNet2DConditionModel` that predicts the noise component ϵ for the target frame, using cross-attention over context tokens at multiple resolutions.

Gradient checkpointing is enabled to reduce memory usage. The model is trained with AdamW and mixed precision (when CUDA is available).

Fig. 6. Context-conditioned UNet: noisy target is denoised while attending to context tokens.

F. Diffusion Schedules

A DDPM schedule is used for training with a scaled linear beta schedule. For inference, DDIM sampling is used to generate frames with fewer steps.

G. Training Objective

The primary loss is mean squared error between predicted and true noise:

$$\mathcal{L}_{\text{MSE}} = \|\epsilon_{\theta}(x_t, t, \text{ctx}) - \epsilon\|_2^2.$$

To improve training stability across timesteps, the loss can be reweighted using a minimum-SNR scheme.

In addition, an ℓ_1 penalty on the predicted clean image x_0 is applied:

$$\mathcal{L}_{x_0} = \|\hat{x}_0 - x_0\|_1,$$

with a fixed weight.

H. Sampling and Rollouts

Qualitative monitoring is performed by periodically saving sample grids that include: the k context frames, the ground-truth target, and the generated next frame. Separate sampling is supported for transition and non-transition windows. Multi-step rollouts are implemented by iteratively appending generated frames to the context window.

a) *Training setup.*: The diffusion model was trained for next-frame prediction using a context-conditioned UNet backbone and a DDPM noise schedule, with DDIM used for accelerated sampling. Training optimizes noise-prediction mean squared error with an additional L_1 regularization term on the reconstructed clean image. The model is conditioned on a fixed window of previous frames and an absolute frame index embedding to provide temporal context. Hyperparameters were selected empirically based on stability and computational constraints, following standard diffusion model practices [2].

Parameter	Value
Context window (k)	7 frames
Image resolution	128×128
Batch size	32
Training epochs	1000
Learning rate	1×10^{-4}
LR scheduler	ReduceLROnPlateau
Diffusion timesteps	600
DDIM sampling steps	150
Time embedding dimension	8
Context embedding dimension	128
L_1 regularization weight	0.1

TABLE III
DIFFUSION MODEL TRAINING HYPERPARAMETERS.

All hyperparameters were selected based on common practice in diffusion-based image generation, as well as preliminary empirical experimentation under computational and time constraints. No classifier guidance or classifier-free guidance was used during training or sampling.

The model was trained to predict only the next frame given the conditioning context, without explicit long-horizon objectives. As a result, temporal consistency beyond short rollouts is not explicitly enforced during optimization.

V. EXPERIMENTS AND RESULTS

This section presents a test-only evaluation of the trained diffusion model. All experiments use a fixed pretrained checkpoint and a frozen configuration to ensure reproducibility.

The evaluation addresses three questions: (i) next-frame prediction accuracy, (ii) behavior under multi-step rollouts, and (iii) performance differences between phase transitions and non-transitions.

A. Evaluation Setup

All experiments are conducted on the held-out test split, defined at the embryo level. The test dataset consists of windowed samples constructed from the silver dataset, using strictly adjacent temporal pairs. Evaluation uses a fixed random seed and deterministic settings.

Inference is performed using DDIM sampling with a fixed number of steps and zero stochasticity. All images are generated at the same resolution used during training.

B. One-Step Prediction

The primary experiment evaluates one-step next-frame prediction under teacher forcing. Given a context of k ground-truth frames, the model generates the frame at $t+1$. Generated frames are compared against the ground-truth target.

Performance is measured using mean squared error (MSE), mean absolute error (MAE), structural similarity index (SSIM), and LPIPS. Metrics are averaged over a fixed number of randomly sampled test windows.

Setting	MSE	MAE	SSIM	LPIPS
One-step (all)	0.145079	0.247196	0.435961	0.116461

TABLE IV
QUANTITATIVE RESULTS FOR ONE-STEP NEXT-FRAME PREDICTION ON THE TEST SET.

C. Transition vs Non-Transition Analysis

To assess model behavior across developmental changes, one-step predictions are separated into *transition* and *non-transition* cases. A transition is defined as a change in morphokinetic phase label between time t and $t+1$.

Results indicate comparable performance between transition and non-transition frames, suggesting that the model does not collapse when predicting across phase boundaries.

Subset	MSE	MAE	SSIM	LPIPS
Transition	0.142403	0.243072	0.435479	0.118123
Non-transition	0.141820	0.244534	0.431187	0.109960

TABLE V
ONE-STEP PREDICTION PERFORMANCE ON TRANSITION AND NON-TRANSITION SAMPLES.

D. Multi-Step Rollout Evaluation

To evaluate temporal stability, the model is tested under free-running rollouts. Starting from a ground-truth context window, the model iteratively generates future frames, which are appended to the context and used as input for subsequent steps.

Metrics are computed independently at each rollout step by comparing generated frames to ground-truth frames at the corresponding time indices. Performance degrades gradually with rollout depth, reflecting error accumulation over time.

Step	MSE	MAE	SSIM	LPIPS
$t+1$	0.144584	0.237566	0.447839	0.086642
$t+2$	0.189093	0.291528	0.357182	0.135496
$t+3$	0.185016	0.292097	0.372274	0.167797
$t+4$	0.193077	0.304039	0.346380	0.195927
$t+5$	0.218165	0.330689	0.302630	0.213073
$t+6$	0.249111	0.364531	0.250075	0.259712
$t+7$	0.228252	0.349119	0.294319	0.256934

TABLE VI
ROLLOUT PERFORMANCE AS A FUNCTION OF PREDICTION HORIZON.

E. Qualitative Results

Qualitative evaluation is performed by visualizing generated frames alongside their conditioning context and ground-truth targets. Both one-step predictions and multi-step rollouts are visualized. Rollout grids highlight the gradual drift that emerges over longer horizons, while preserving overall embryo structure in early steps.

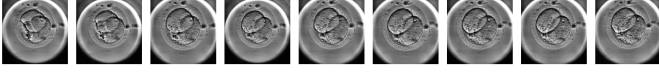


Fig. 7. Qualitative one-step prediction examples: context frames, ground truth, and generated frame.

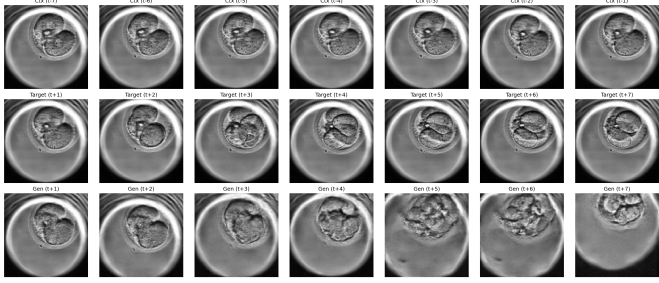


Fig. 8. Qualitative multi-step rollout examples comparing generated and ground-truth frames.

VI. DISCUSSION AND NEXT STEPS

A. Discussion

The experimental results indicate that the proposed context-conditioned diffusion model is able to generate visually plausible next-frame predictions in human embryo time-lapse sequences. For short-term transitions, the model preserves coarse embryo structure and maintains temporal continuity with the conditioning frames. These results demonstrate the feasibility of diffusion-based approaches for modeling embryo morphokinetic dynamics at the frame level.

However, the model is not sufficiently accurate to reliably ensure correct morphological consistency across transitions, particularly when evaluated over longer temporal horizons. Multi-step rollouts reveal noticeable error accumulation, manifested as structural drift, deformation, and loss of fine morphological detail. While this behavior is expected in autoregressive generation, it indicates that the current model does not robustly capture the underlying developmental dynamics required for long-term prediction.

Overall, the presented results should be interpreted as a baseline. The current model provides a proof of concept rather than a fully reliable predictive system. Achieving biologically consistent long rollouts remains challenging and would require further architectural refinements, stronger supervision, and more extensive training.

B. Limitations

Several limitations affect the current approach. The model is trained and evaluated using single-plane grayscale imagery,

which restricts its ability to capture three-dimensional embryo structure. Morphological consistency is not explicitly enforced during training, and evaluation relies primarily on pixel-level and perceptual metrics that do not fully reflect biological correctness.

In addition, although phase annotations are available, they are not directly incorporated into the diffusion training objective. As a result, the model does not explicitly reason about discrete developmental states, which likely contributes to inconsistencies across phase transitions and long rollouts.

Finally, due to time constraints within the thesis period, training duration, hyperparameter exploration, and large-scale architectural experimentation were limited. These constraints directly impact the achievable performance and stability of the model.

C. Next Steps

Future work will focus on improving morphological consistency and temporal stability. Planned directions include incorporating explicit phase conditioning or phase-aware objectives, exploring alternative loss formulations that better constrain long-term evolution, and investigating curriculum or multi-horizon training strategies.

Additional improvements may include extended training, larger models, and multi-plane or inputs to better capture embryo structure. The next period will aim to move beyond baseline performance toward more accurate and biologically consistent modeling of embryo development over extended time horizons.

REFERENCES

- [1] T. Gomez *et al.*, “A time-lapse embryo dataset for morphokinetic parameter prediction,” *Data in Brief*, vol. 42, 2022.
- [2] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020.